The Wage Game: A Case Study

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An Experiment on Efficiency Wages

I. Introduction

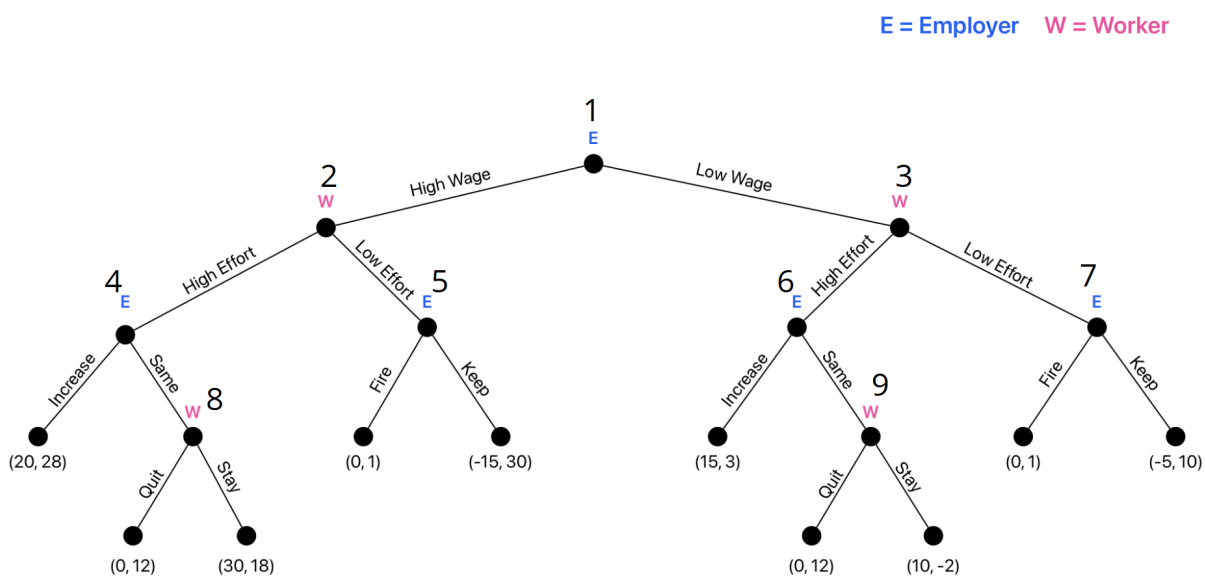
In the paper "Bargaining and Repeated Games" by Jonathan Levin (2002), the concept of efficiency wages is discussed in the context of repeated games. Efficiency wages are wages clearing the market rate offered to an employee to incentivize higher productivity. For our experiment, we chose to study the broad category of sequential games—the theories and methodologies behind successive decisions in a given scenario. Particularly, we were interested in the relationship between employers and employees in a firm where efficiency wages had the potential to be offered—how would employees react, and is it worth the investment on the part of the firm? We sought to model our experiment after this scenario and compare it to theory.

II. The Model

For our experiment we created a two-player game—one acting as the employer, who represents the firm, and the other being the worker. Matchings were random and fixed throughout the entirety of the game, to ensure stability in the results and capture the effect of such repeated interactions between parties. The first mover in each round is the employer, who decides to offer an initial wage of High (W_H) or Low (W_L). W_H is valued at 30 while W_L is 10. Knowing their wage, the worker next chooses their level of effort based on this wage: High Effort, with a value of wage*2, or Low Effort, with a value of wage*0.5. If the chosen effort is low, the employer is then faced with the decision of firing the employee, which carries a penalty for both employer and employee, or keeping them at a detriment to the firm. If the chosen effort is high however, the employer then has the choice to either reward the employee with a wage increase (W_M) or keep the same wage. A wage increase from W_H is then valued at 40, while a raise from W_L is then brought to an average wage (W_A) of 15. Additionally, a raise is associated

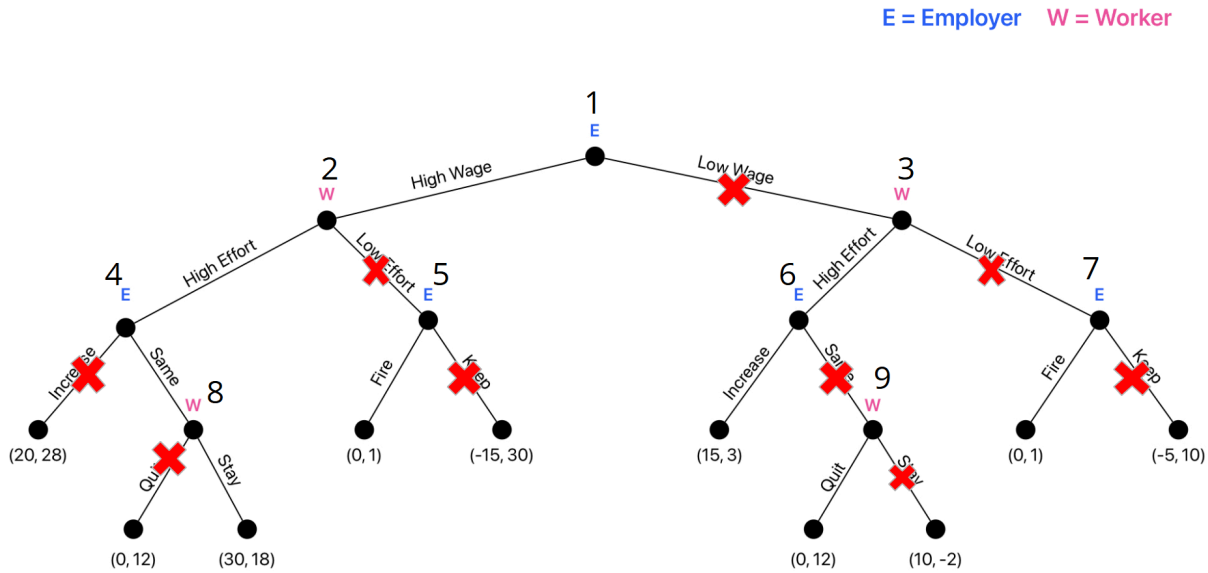
with an increase in productivity. Finally, if the employee has chosen a high effort regardless of wage but is not offered a raise, then they have the choice to quit the firm or stay. The cost of staying is quantified in the loss of leisure time.

Payoffs for the employer are calculated as the worker's effort level minus their paid wage. On the other hand, the employee's payoff is calculated as their wage minus leisure cost for high effort, and is equal to their wage for low effort. Consequently, if they quit, their payoff is equal to their leisure (12), and if fired, is leisure minus their assumed reputation damage (11). Moreover, if the worker is originally offered W_H, has high effort, and is given a wage increase to W_M, then their effort is only equal to wage*1.5. This can be attributed to a diminishing return on wage, and limits the maximum value of productivity. The sequence of decisions with associated payoffs are illustrated in the table below.



III. Solving for Nash Equilibrium

As with most sequential game models, our model can be solved using backward induction. As the last mover, we address the employee's decisions first. In the case of quitting or staying, which always comes about under a high effort (wage minus leisure) and no raise scenario, it depends on the initially offered wage. For W_H, the employee will choose to stay as their payoff is 18 compared to 12 for quitting. For W_L, the employee will choose to quit as their payoff is 12 compared to -2. Knowing this, the employer then makes the decision to offer the employee a raise or not. In the case of W_H, the employer will not offer a raise to W_M for the employee knowing they would stay regardless, as their payoff (effort minus wage) is higher for the lower wage with the same effort, 30 compared to 20. For W_L, however, the employer would offer the employee a raise to W_A, as the cost of the raise is lower than the cost of training a new employee, 15 compared to 0. Additionally, in the case of low effort, the employer will always choose to fire the employee, as the cost of training a new employee is lower than the cost of lost productivity. Knowing this, the employee will always choose a high effort when acting rationally, as their payoff for high effort is always higher than the low effort and get fired pathway, regardless of initial wage. Finally, at the initial decision node, the employer will always choose a high initial wage, knowing the employee will give a high effort regardless, and that the employer will not offer a raise. This payoff of 30 for the employer is higher than the expected payoff of 15 if the low wage is initially offered. Thus, if the players follow the Subgame Perfect Nash Equilibrium (SPNE) and do not deviate from their best choice at each node, the optimal outcome for our model is High Wage, High Effort, No Raise, and Stay, with a payoff of 30 for the employer and 18 for the employee. A trimmed game tree depicts this logic below.



IV. The Experiment

The experiment featured random matching and no communication between participants, and participants were not given a game tree but were informed of how to calculate the payoffs prior to beginning. The experiment was designed to test the tendency of participants to make efficient responses, or rather, the experiment is testing wage efficiency. Across four rounds, it was hoped that participants would formulate a game tree and utilize backward induction in the manner described above, and act accordingly. The data we collected from the experiment consisted of 52 total games, with 13 pairings in four rounds. Out of ten potential outcomes, three of them constituted a majority 33 of all 52 games, or roughly 64%. These were all of the W_H, high effort variations. This indicates that our experiment does predict these first two decisions as most likely, which is in-line with our model. Furthermore, it supports the efficiency wage theory that a wage higher than the industry average may be used by employers to incentivize greater productivity. This is further supplemented by the fact that 33% of games consisted of W_H, high

effort, and an additional wage increase despite the lack of increase in nominal productivity. However, 28 of these particular 33 games were from the wage increase and quitting outcomes, that are not in line with the model which had no increase and staying as the Nash equilibrium. This could be attributed to spite of employees, in which participants would rather bear the cost of quitting in hopes of finding an employer where their effort would be rewarded. Using results from Round 1 as a one-shot game, it can be seen that the Nash equilibrium was reached three out of 13 rounds, 23% of the time, a much higher rate than in the ensuing rounds. This is an indication that the repeated aspect of our game had an effect which produced insignificant evidence for our optimal outcome beyond the one shot model.

Data

	WH, EH, N, Q	**WH, EH, N, S**	WH, EH, I	**WH, EL, F**	WH, EL, K	**WL, EH, N, Q**	WL, EH, N, S	**WL, EH, I**	WL, EL, F	WL, EL, K	
Payoff key: (Employer, Worker)	(0,12)	**(30,18)**	(20,28)	**(0 ,1)**	(-15,30)	**(0 ,12)**	(10,-2)	**(15,3)**	(0,1)	(-5,10)	
Round 1	3	**3**	2	**0**	1	**0**	1	**0**	1	2	13
	23%	**23%**	15%	**0%**	8%	**0%**	8%	**0%**	8%	15%	
Round 2	3	**0**	7	**1**	0	**0**	0	**1**	0	1	13
	23%	**0%**	54%	**8%**	0%	**0%**	0%	**8%**	0%	8%	
Round 3	4	**1**	4	**0**	1	**1**	0	**1**	1	1	13
	31%	**8%**	31%	**0%**	8%	**8%**	0%	**8%**	8%	8%	
Round 4	1	**1**	4	**2**	2	**0**	0	**2**	1	0	13
	8%	**8%**	31%	**15%**	15%	**0%**	0%	**15%**	8%	0%	
Total rounds	11	**5**	17	**3**	4	**1**	1	**4**	3	4	52
	21%	**10%**	33%	**6%**	8%	**2%**	2%	**8%**	6%	8%	

V. Results — Theoretical vs. Experimental Disparities

The main culprits for the lack of correlation with the theory can be attributed to emotion. The experiment was a proxy for the real-world frustration often felt regarding differences between wages and effort and offered an easy way for participants to exploit these feelings

through virtually consequenceless actions. Quitting is not a rational decision when the other option is to stay and still receive an above average wage and a higher overall payoff. But if logic falls by the wayside, then the employee feels just quitting and leaving their employer worse off. Similarly, the other most common outcome was for the employer to increase wage after W_H and high effort. In a real-world scenario, this would most likely not happen as the employer looks to maximize profit in the short run. In the experiment however, logic is once again thrown to the wind as employers looked to potentially reward their employees despite no increase in the firm productivity in the short term. This could, however, yield long term benefits like firm loyalty, partially captured by consistency across all rounds of the experiment for certain pairings. In all, our model and theory were mostly accurate, with the statistical differences being easily explainable.

VI. Conclusion

While insightful, our experiment was not without its flaws. Importantly, we believe there were too few rounds. While the results did not converge to the Nash Equilibrium over the four rounds, they showed consistency in the later rounds, allowing the error to be quantified and explained by variable omission. The small number of rounds and limited experiment time also potentially did not give participants enough time to draw and utilize a full and accurate game tree. If further research were to be done, the experiment should have more rounds and a variable to control for emotional response. Future research could also look into the long-term effects of efficiency wages on the labor, wage, and industry markets.

References

Levin, Jonathan. "Bargaining and Repeated Games." *Web.stanford.edu*, Stanford University Press, Feb. 2002, web.stanford.edu/~jdlevin/Econ%20203/RepeatedGames.pdf. Accessed 3 Dec. 2024.